

NATIONAL JUNIOR COLLEGE

Higher 1



CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION NO.

BIOLOGY

8875/02

JC2 Preliminary Examinations
Paper 2

22 Sept 2008

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on the writing paper that you write the essay on.

Write in dark blue or black pen.

Answer **all** questions in **Section A** and **choose one** essay question to answer from **Section B**.

At the end of the exam, staple your essay to the back of the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY	
1	
2	
3	
4	
5	
6/7	
TOTAL (/67)	

This paper consists of **18** printed pages including cover page.

SECTION A: 80 MARKS

Answer all questions.

Question 1

Gastric proton/potassium-ATPase (H^+/K -ATPase) is the major enzyme involved in gastric acid secretion in the stomach. Recognizing the calming effects that tea can have on an upset stomach, biochemists quickly showed that the tannic acid found in tea is a potent inhibitor of this enzyme. Gastric proton/potassium-ATPase (H^+/K -ATPase) has two substrates, potassium ion (K^+) and ATP. (K_m is substrate concentration at $\frac{1}{2} V_{max}$)

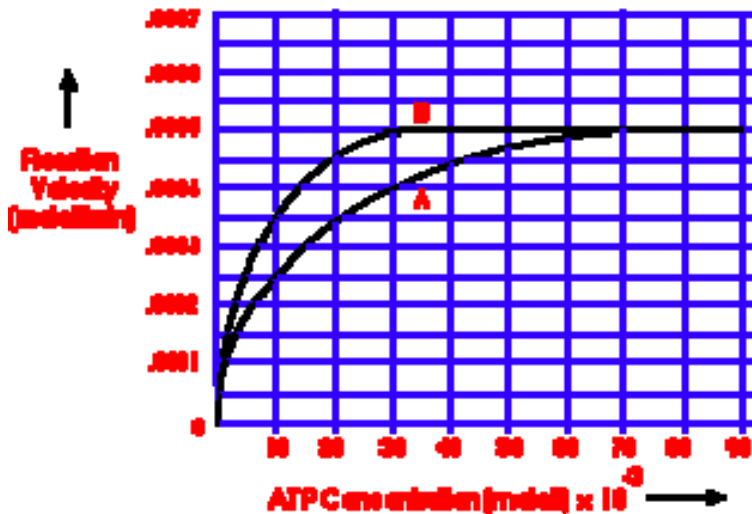


Figure 1.1 Reaction velocity-substrate concentration curves for Gastric proton/potassium-ATPase (H^+/K -ATPase) with unlimiting levels of K^+ but varying levels of ATP. A: with tannic acid; B: without tannic acid.

(a) What is the minimum ATP concentration needed to saturate the enzyme in this system without tannic acid?

[1]

(b) By comparing K_m values based on the graphs in figure 1, explain which treatment would result in the higher affinity of the enzyme for the substrate?

[2]

- (c) i) Name the type of inhibition that is involved above and provide evidences for your choice.

[2]

- ii) With reference to the structure of the active site, suggest the possible mechanism of inhibition and why the inhibitor cannot be broken down by the enzyme.

[2]

- (d) Describe two ways in which enzymes interact with substrates to specifically increase the likelihood of a reaction.

[2]

- (e) Giving an example in each, describe one main difference between prosthetic groups and coenzymes that are involved in enzymatic reactions.

[2]

[Total: 11 marks]

Question 2

A cross was made between a pure breeding round-leafed, tall plant and a oval-leafed, dwarf plant produced all round-leafed, tall plants.

(i) Using this information determine which alleles are dominant and which are recessive.

[2]

*In **another** cross between a round-leafed, tall plant and a round-leafed dwarf plant following offspring were produced:*

*121 round-leafed, tall plants
124 round-leafed, dwarf plants
42 oval-leafed, tall plants
37 oval-leafed, dwarf plants*

Using appropriate symbols, deduce the genotypes of the 2 parents and explain how you arrive at your answer

(ii) Symbols:

[1]

(iii) Genotypes of the 2 parents

[2]

(iv) Explanation:

[3]

[Total: 8 marks]

Question 3

a) What are the main differences in the structure and organization of prokaryotic and eukaryotic genomes?

[4]

b) Describe two functions of repetitive sequences.

[2]

[Total: 6 marks]

Question 4

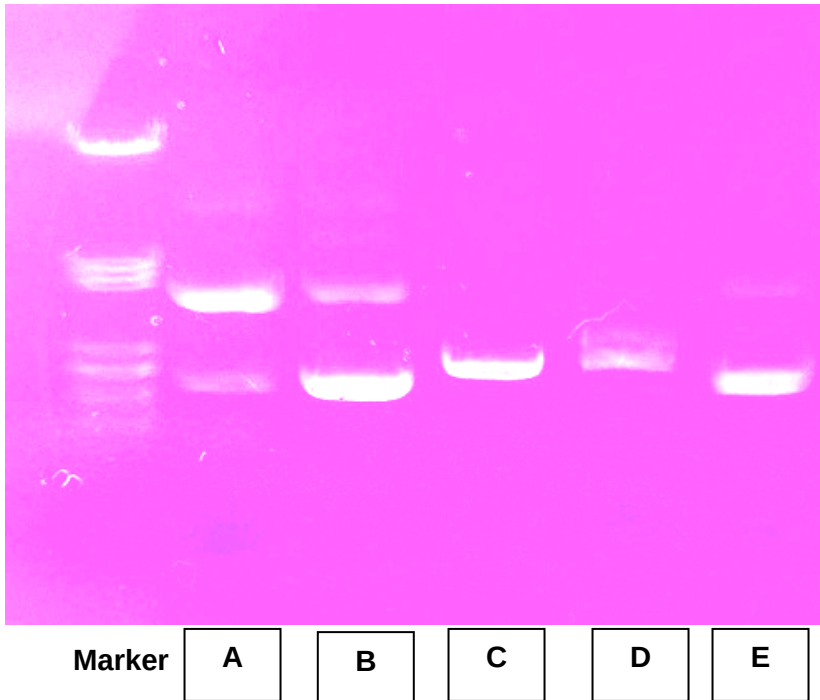


Fig. 5.1 shows a gel electrophoresis result of DNA obtained from similar looking flagella obtained from 5 species of marine bacteria. They were all treated with the same restriction enzyme. Species A to E were each retrieved from different oceans of the world.

- (a) i) Name the two species that are most likely to evolve flagella similar to A, as a result of convergent evolution.

[1]

- ii) Explain how the method of analysis has supported your choice of answer.

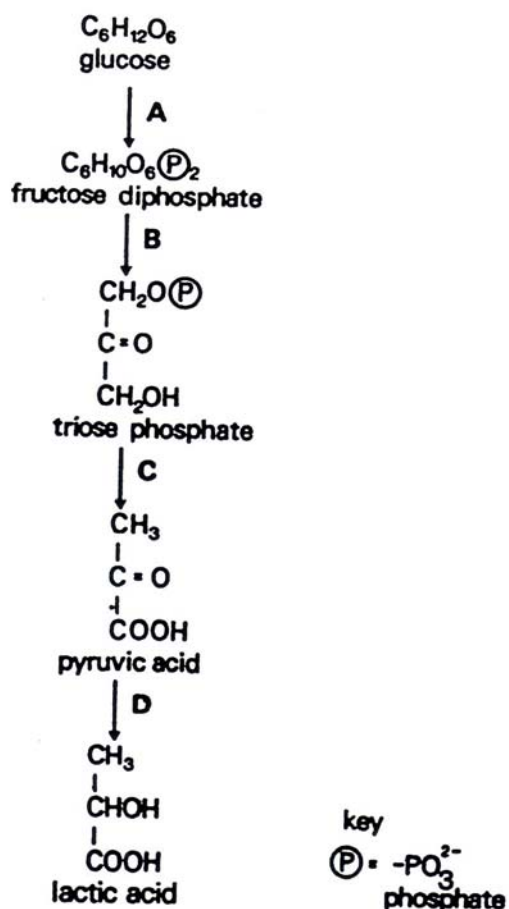
[2]

d) Outline the processes and principle of Agarose Gel Electrophoresis.

[Total: 11 marks]

Question 5

In anaerobic respiration in muscle tissue the conversion of glucose to lactic acid can be represented as a four stage process as follows:



- (a) Name the four stages in terms of the chemical processes occurring.

A

B

C

D

[2]

- (b) At what stage would you expect hydrogen transfer to NAD to occur?

[1]

- (c) Is there a net change in the concentrations of NAD and NADH₂ in the whole process? Explain your answer.

[3]

- (d) If one molecule of glucose is converted to two molecules of lactic acid, what is the net synthesis of ATP? Explain your answer.

[1]

Explanation:

[2]

- (e) In what ways is the ethanol formation different from that of lactate?

[2]

[Total: 11 marks]

SECTION B: 20 MARKS
Choose **EITHER** Question 8 **OR** Question 9.

Write your answers to this question on a separate piece of writing paper.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b), etc., as indicated in the question.

Question 6

- a) What are the advantages and disadvantages of the use of PCR? [6]
- b) Contrast the features and uses of totipotent, pluripotent and multipotent stem cells. [6]
- c) Outline the procedure to produce the human growth hormone (hGH) from bacterial cells. [8]

Question 7

- a) Describe in detail the structure of DNA. [12]
- b) Explain, in general, the main events of transcription. [8]

--- The End ---